

Lab Assignment 3

Object-Oriented Programming

Branch

B.E., 2nd Year – ENC



Submitted By:

Kaushtubh Kumar

Roll No. 1024150191

Batch: 2023

Q1 Write a program to implement (a) pointer to an object (b) this pointer. Practice both " (dot operator) and '->' (arrow operator).

CODE
<pre>#include <iostream> using namespace std; class ClassB; class ClassA { int valA; public: ClassA(int v) { valA = v; } friend void swapValues(ClassA&,ClassB&); void show() { cout << "ClassA: " << valA << endl; } }; class ClassB { int valB; public: ClassB(int v) : valB(v) {} friend void swapValues(ClassA &, ClassB &); void show() { cout << "ClassB: " << valB << endl; } }; void swapValues(ClassA &a, ClassB &b) { int temp = a.valA; a.valA = b.valB; b.valB = temp; } int main() { ClassA obj1(10); ClassB obj2(20); cout<<"Before swap:"<<endl; obj1.show(); obj2.show();</pre>

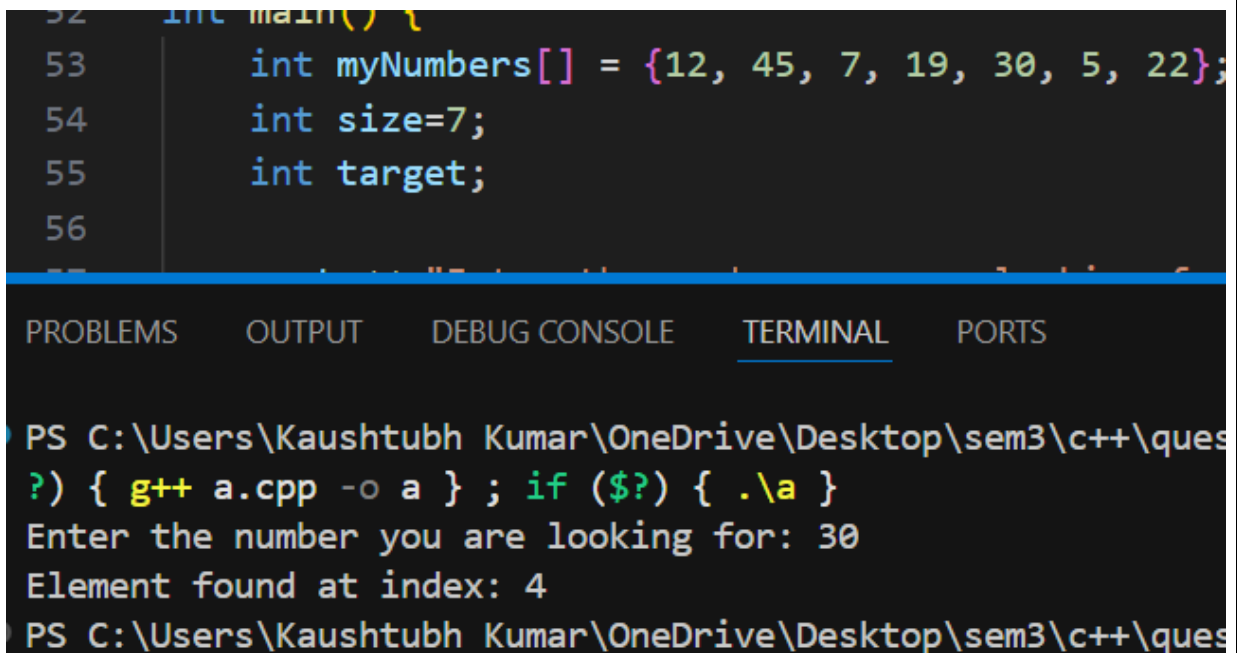
```

swapValues(obj1, obj2);

cout<<"After swap:"<<endl;
obj1.show();
obj2.show();
return 0;
}

```

TESTCASE



The screenshot shows a C++ IDE with a dark theme. The top part displays C++ code for a program that finds a target in an array. The bottom part shows the terminal output where the user enters '30' and the program outputs 'Element found at index: 4'.

```

52 int main() {
53     int myNumbers[] = {12, 45, 7, 19, 30, 5, 22};
54     int size=7;
55     int target;
56
57     // ... (code continues) ...

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\Kaushtubh Kumar\OneDrive\Desktop\sem3\c++\ques
?) { g++ a.cpp -o a } ; if ($?) { .\a }
Enter the number you are looking for: 30
Element found at index: 4
PS C:\Users\Kaushtubh Kumar\OneDrive\Desktop\sem3\c++\ques

```

Q2. Write a program to swap private values of two classes using a friend function.

CODE

```

#include <iostream>
using namespace std;

class ClassB; // Forward declaration

class ClassA {
    int a;
public:

```

```

    ClassA(int v) : a(v) {}
    void show() { cout << "A: " << a << endl; }
    friend void swap(ClassA &objA, ClassB &objB);
};

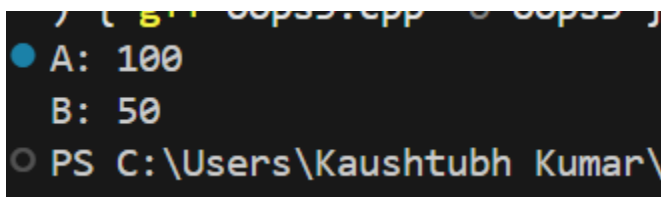
class ClassB {
    int b;
public:
    ClassB(int v) : b(v) {}
    void show() { cout << "B: " << b << endl; }
    friend void swap(ClassA &objA, ClassB &objB);
};

void swap(ClassA &objA, ClassB &objB) {
    int temp = objA.a;
    objA.a = objB.b;
    objB.b = temp;
}

int main() {
    ClassA x(50);
    ClassB y(100);
    swap(x, y);
    x.show(); y.show();
    return 0;
}

```

TESTCASE



```

PS C:\Users\Kaushtubh Kumar\
A: 100
B: 50

```

Q3. Write a program to add data objects of two different classes using friend functions.

CODE

```
#include <iostream>
```

```

using namespace std;

class ClassY;

class ClassX {
    int data;
public:
    ClassX(int d) : data(d) {}
    friend void sum(ClassX, ClassY);
};

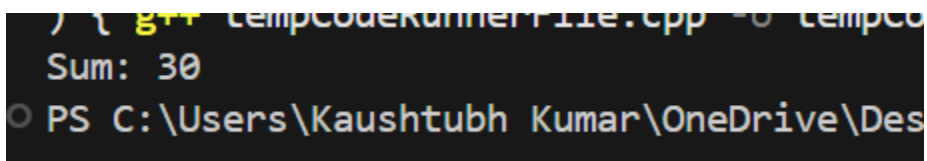
class ClassY {
    int data;
public:
    ClassY(int d) : data(d) {}
    friend void sum(ClassX, ClassY);
};

void sum(ClassX obj1, ClassY obj2) {
    cout << "Sum: " << obj1.data + obj2.data << endl;
}

int main() {
    ClassX a(10);
    ClassY b(20);
    sum(a, b);
    return 0;
}

```

TESTCASE



```

) { g++ tempCodeKunner11e.cpp -o tempco
Sum: 30
PS C:\Users\Kaushtubh Kumar\OneDrive\Des

```

Q4. Write a program to demonstrate the working of friend class.

CODE

```

#include <iostream>
using namespace std;

```

```

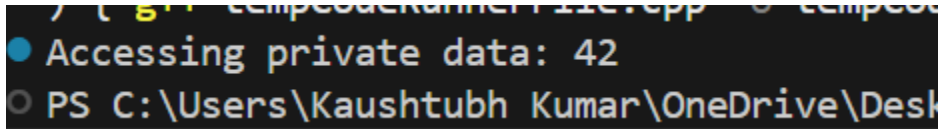
class Base {
    int secret = 42;
    friend class Auditor;
};

class Auditor {
public:
    void reveal(Base &b) {
        cout << "Accessing private data: " << b.secret <<
endl;
    }
};

int main() {
    Base b;
    Auditor a;
    a.reveal(b);
    return 0;
}

```

TESTCASE



```

C:\Users\Kaushtubh Kumar\OneDrive\Desktop> g++ tempcodekunner11c.cpp -o tempcodekunner11c.exe
Accessing private data: 42
PS C:\Users\Kaushtubh Kumar\OneDrive\Desktop>

```

Q5 Write a program using Array of Objects to display area of multiple rectangles.

CODE

```

#include <iostream>
using namespace std;

class Rectangle {
    float l, b;
public:
    void setData(float length, float breadth) {
        l = length; b = breadth;
    }
}

```

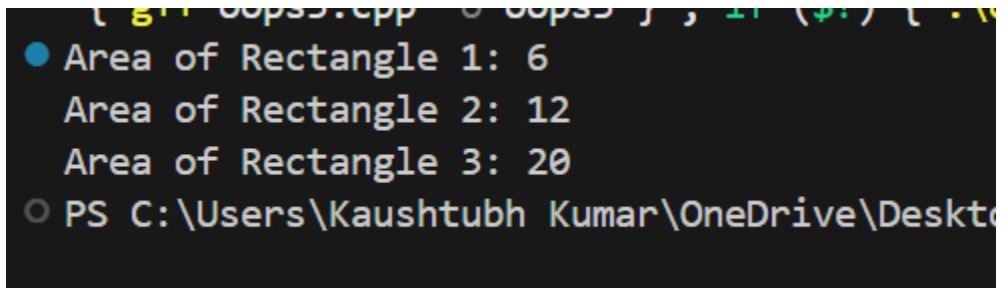
```

float getArea() { return l * b; }
};

int main() {
    Rectangle rects[3];
    for(int i = 0; i < 3; i++) {
        rects[i].setData(i + 2, i + 3);
        cout << "Area of Rectangle " << i+1 << ": " <<
rects[i].getArea() << endl;
    }
    return 0;
}

```

TESTCASE



```

C++ g++ gpp3.cpp -o gpp3 ; if ($?) { .\gpp3.exe
● Area of Rectangle 1: 6
  Area of Rectangle 2: 12
  Area of Rectangle 3: 20
○ PS C:\Users\Kaushtubh Kumar\OneDrive\Desktop

```

Q6. Write a program to define function cube() as inline for calculating cube of a number.

CODE

```

#include <iostream>
using namespace std;

inline int cube(int n) {
    return n * n * n;
}

int main() {
    int num = 5;
    cout << "Cube of " << num << " is " << cube(num) <<
endl;
    return 0;
}

```

```
}
```

TESTCASE

```
{ g++ oops3.cpp -o oops3 } ; if ($?) {  
    Cube of 5 is 125  
PS C:\Users\Kaushtubh Kumar\OneDrive\Desktop
```

Q7 Write a program to pass an object as an argument and return the object from a function.

(a) Use pass-by-value

(b) Use pass-by-address

CODE

```
#include <iostream>  
using namespace std;  
  
class Box {  
public:  
    int value;  
    Box(int v) : value(v) {}  
};  
  
Box processByValue(Box b) {  
    b.value += 10;  
    return b;  
}  
  
(Box* processByAddress(Box *b) {  
    b->value += 20;  
    return b;  
}
```



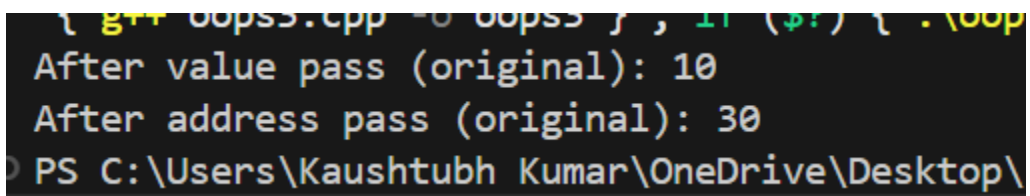
```
int main() {
    Box myBox(10);

    Box result1 = processByValue(myBox);
    cout << "After value pass (original): " << myBox.value
    << endl; // Remains 10

    Box* result2 = processByAddress(&myBox);
    cout << "After address pass (original): " <<
    myBox.value << endl; // Changes to 30

    return 0;
}
```

TESTCASE



```
{ g++ oopss.cpp -o oopss } ; if ($?) { .\oop
After value pass (original): 10
After address pass (original): 30
PS C:\Users\Kaushtubh Kumar\OneDrive\Desktop\
```